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THE "BURNING SHIP" AND ITS QUASI-JULIA SETS

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Abstract—By manipulating the imaginary part of the complex-analytic quadratic equation, we obtain, by iteration, a set that we call the "burning ship" because of its appearance. It is an analogue to the Mandelbrot set (M-set). Its nonanalytic "quasi"-Julia sets yield surprizing graphical shapes.

1. INTRODUCTION

The real and imaginary components of the complex quadratic equation are

$$x_{n+1} = x_n^2 - y_n^2 - c_x \quad \text{and}$$

$$y_{n+1} = 2x_n y_n - c_y. \qquad (1)$$

Iterating Eq. (1) in the c-plane (parameter space) yields for $x_o = 0$ and $y_o = 0$ under the usual manipulations [1, 2], the Mandelbrot set (M-set).

We arbitrarily replace some elements of the right-hand side of Eq. (1) by their absolute values. Thereby, the analyticity of this equation breaks down. A first such case was considered in [3]. Here we present results for the following equation:

$$x_{n+1} = x_n^2 - y_n^2 - c_x \quad \text{and}$$

$$y_{n+1} = 2|x_n y_n| - c_y \qquad (2)$$

Its quasi-Julia sets (QJS's) apply in the x, y plane (state space). The word "quasi" is used because the present maps are not analytic, that is, they do *not* obey the Cauchy-Riemann conditions

$$\partial f_1(x, y)/\partial x = \partial f_2(x, y)/\partial y$$

$$\partial f_1(x, y)/\partial y = -\partial f_2(x, y)/\partial x$$

2. STRUCTURE AND PROPERTIES OF THE "BURNING SHIP"

Figure 1 shows an analogue to the M-set in parameter space. We arbitrarily put $x_o = y_o = 0$. The main body in this figure is called the burning ship.

1. The length of the main body in the c_x-direction is the same as for the M-set. Along the c_x-axis (antenna region of the M-set), at identical positions, occupied by the small M-sets, "mini-ships" are located.
2. The main body can be divided into five regions that each exhibit different properties (Fig. 1):

   Region 1 of Fig. 1 is below the "water line" of the ship, $c_y < 0$. The range $-.5 < c_x < -.25$ we call the "bump." Its boundary is slightly irregular; in its front range, $-.43 < c_x < -.39$ and $-.2 < c_y < -.17$, the boundary is actually dust-like.

   Region 2 also lies below the water line to the right of Region 1, extending up to $c_x = 1.8$. These "keels," both of the main ship and those of the mini ships, are mostly smooth. Some intervals (in the main keel at $.5 < c_x < .65$, $1.15 < c_x < 1.25$, and $1.35 < c_x < 1.4$), and corresponding regions of the mini-ships, show beard-like structures (Fig. 2).

   Region 3 lies above the water line for $c_x > 1.46$. It also includes all upper parts of the mini-ships. There are Eiffel Tower-like superstructures at low iteration number limits of about 50 (Fig. 3a, b, c).

   Region 4 lies to the left of Region 3, comprizing the upper—"burning"—part of the ship. Its boundary is "dusty" in contrast to the corresponding parts of the mini-ships. Within the dust dumb-bell-like voids of all sizes are randomly distributed (Fig. 4).

   Region 5 is the "bow" of the burning ship, lying at $-.8 < c_x < -.5$ and $0 > c_y > 1.1$. Its boundary is smooth (Fig. 1, left hand edge).

3. CORRESPONDING QUASI-JULIA SETS (QJS's)

When iterating Eq. (2), with fixed p_x and p_y values replacing the previously variable c_x and c_y values, we obtain QJS's in the x, y plane. Bearing close to the border of ships' bodies, tantalizing pictures can be obtained.

- All QJS's along the x-axis, i.e. with differing p_x-values but $p_y = 0$, are identical with the Julia sets of the M-set. This is because the second variable of Eq. (2) is squared immediately at each iteration step, within the first part of Eq. (2).
- All pairs of p-values that correspond to smooth boundary points of the burning ship in Fig. 1 yield smooth QJS's. They explode when the borders of the ship are crossed from the inside, similarly as this is known from the Julia sets of the M-set. (A difference is that the coincidence is not perfect this time because different x_o, y_o pairs yield slightly different canonical boundaries for the ship.) QJS's that correspond to the beard-like intervals of Region 2 exhibit highly esthetical, self-similar fractal ornaments (Figs. 5, 6, 7).
- Eiffel Tower-like superstructures above the Region-3 ships gives rise to corresponding features in the QJS's (Fig. 8a, b, c). Its ordered Eiffel-Tower superstructures are similar to structures reported in [4] for the simplest nonanalytic Julia-like set.
- Corresponding to the randomly distributed dumbbell void structures of Region 4, we find QJS's that have

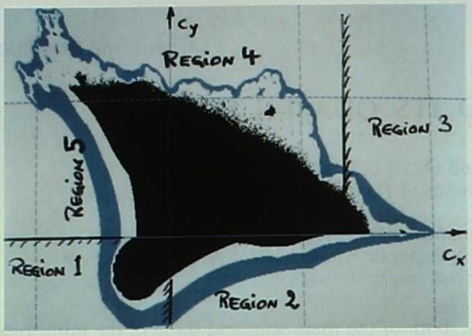

Fig. 1. Computer scan of the "burning ship". Eq. (2) with $x_o = y_o = 0$; maximum iteration number = 250; explosion threshold $x^2 + y^2 = 200$; coordinates: $-1.25 < c_x < 2.25$; $-.75 < c_y < 1.75$.

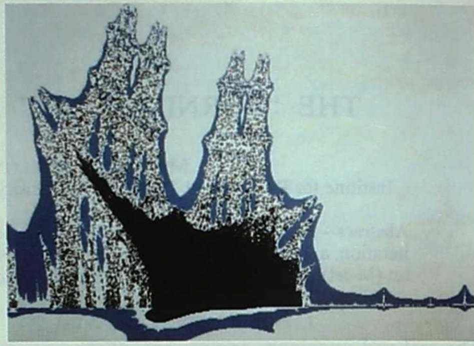

Fig. 3b. The largest of the mini ships, antenna region. Note the Eiffel Tower superstructures. Maximum iteration number = 50; explosion threshold = 50; $1.7 < c_x < 1.8$; $-.01 < c_y < .1$.

Fig. 2. "Beard-like" structures (arrows) in the keels in Region 3. Vertical blow-up; $.3 < c_x < 1.8$; $-.01 < c_y < .1$.

Fig. 3c. Detail of Fig. 3b, main left hand Eiffel Tower; $1.74 < c_x < 1.75$; $-.043 < c_y < .05$.

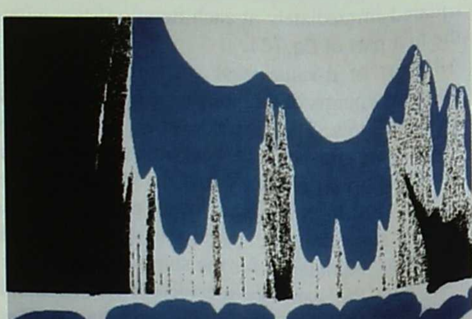

Fig. 3a. Eiffel Tower-like superstructures above the "water line" near the rear end of the ship (main body) and the mini ships of the antenna region (Region 3). $1.4 < c_x < 1.75$; $-.01 < c_y < .1$.

Fig. 4. "Dusty" area of Region 4, showing dumbbell-like voids and randomly distributed mini-ships. Maximum iteration number = 250; explosion threshold = 250; $.6 < c_x < 1.01$; $.6 < c_y < 1.01$.

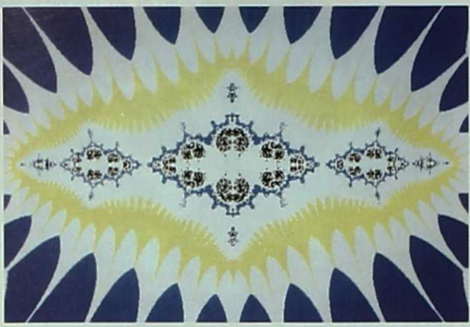

Fig. 5. Exploding quasi-Julia set (QJS) of Region 2. Note the dust-like self-similar rhomb-shaped ornamental structures: $P_x = 1.35$, $P_y = -.0022$. Maximum iteration number = 200; explosion threshold = 200; $-.6 < x < .6$; $-.4 < y < .4$.

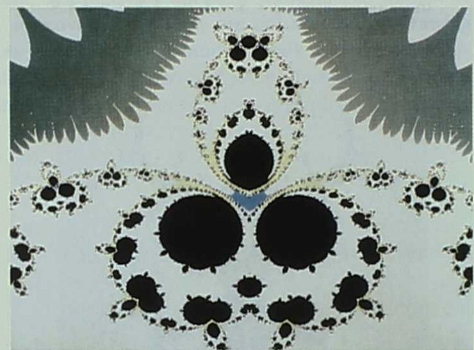

Fig. 6. Detail from an exploding QJS in region 2. $P_x = .3$, $P_y = -.13$. Maximum iteration number = 150; explosion threshold = 8; $-.18 < x < .18$; $.57 < y < .89$.

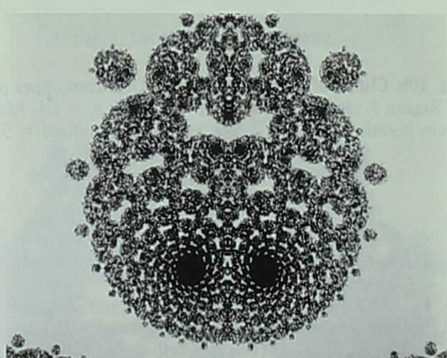

Fig. 7. Detail from an other exploding QJS from Region 2. $P_x = .5$, $P_y = -.07$. Maximum iteration number = 150; explosion threshold = 8; $-.115 < x < .115$; $.675 < y < .860$.

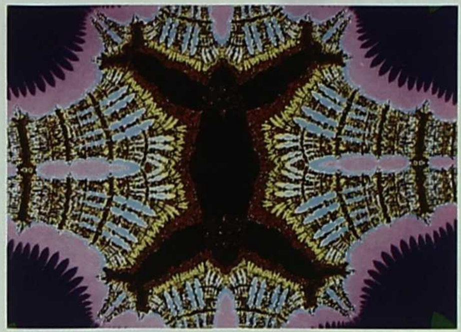

Fig. 8a. "Gorilla" in branches of Eiffel Tower. Detail from Region-3 mini-ship. $P_x = 1.743$, $P_y = .02$. Maximum iteration number = 120; explosion threshold = 8; $-.33 < x < .33$; $-.25 < y < .25$.

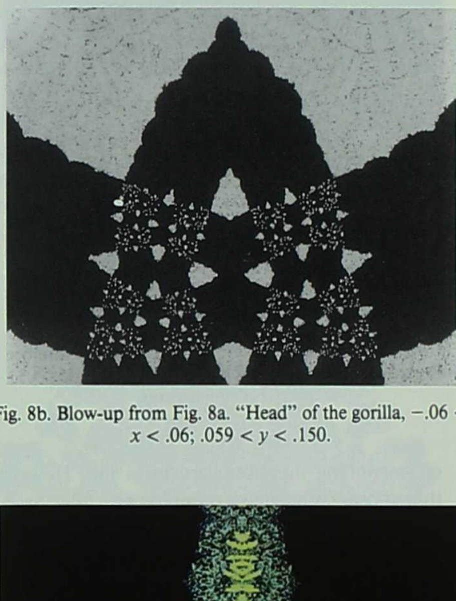

Fig. 8b. Blow-up from Fig. 8a. "Head" of the gorilla, $-.06 < x < .06$; $.059 < y < .150$.

Fig. 8c. Detail from Fig. 8b. "Hole" in the center of the gorilla's chest. $-.0052 < x < -.005$; $.0610 < y < .0750$.

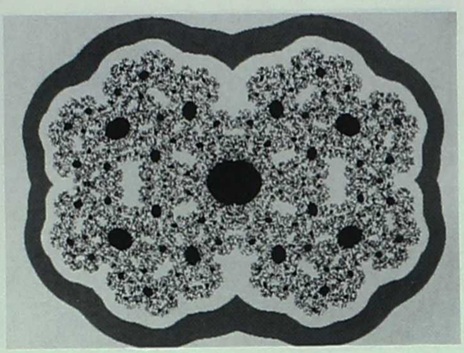

Fig. 9a. "Alzheimer Brain." QJS from Region 4. $p_x = .75$, $p_y = .9$. Maximum iteration number = 120; explosion threshold = 120; $-3 < x < 3$; $-2.2 < y < 2.2$.

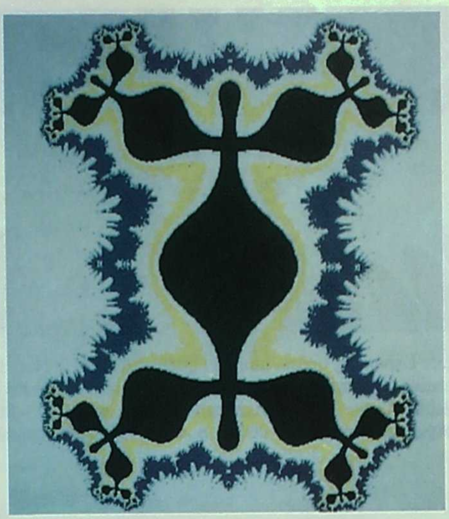

Fig. 10a. Smooth QJS from Region 5, a "Frog". $p_x = -.45$; $p_y = .70$. Maximum iteration number = 150; explosion threshold = 8; $-1.2 < x < 1.2$; $-1.3 < y < 1.3$.

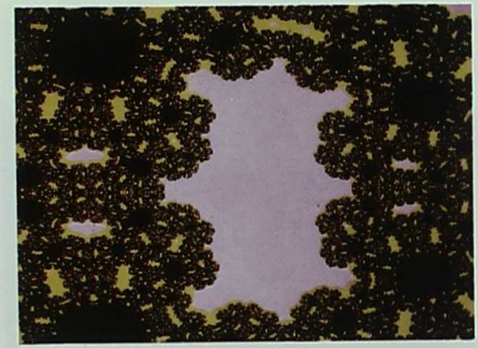

Fig. 9b. Detail from Fig. 9a; maximum iteration number = 400; explosion threshold = 200; $-1.15 < x < -.55$; $-.31 < y < .31$. Compare text.

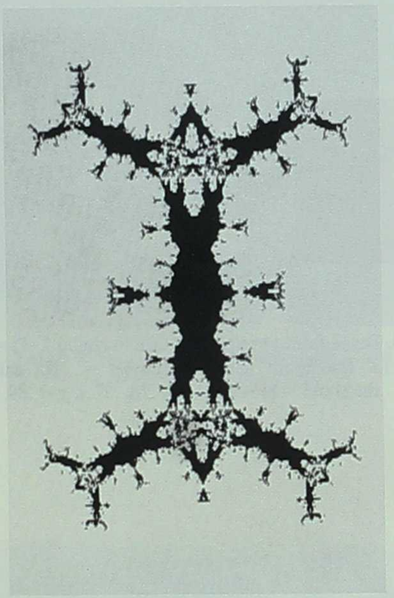

Fig. 10b. Clutch of selfsimilar "Frogs." QJS from upper part of Region 5, close to region 4. $p_x = -.7387$; $p_y = 1.1$. Maximum iteration number = 500; explosion threshold = 500; $-1 < x < 1$; $-2.2 < y < 2.2$.

corresponding dumbbell structures too. However, they appear completely symmetric now (Fig. 9a, b). Note that even the outer circumference of these QJS's reproduces the (inverse) shape of these void structures (Fig. 9a). Those dumbbell voids are, thus, selfsimilar.

- The smooth structures of the Region 5 ("bow") yield smooth QJS's at corresponding p-values. An example is Fig. 10a. A QJS from the transient zone between Regions 4 and 5 is shown in Fig. 10b.

4. CONCLUDING REMARKS

Nonanalytic, noninvertible maps of two variables exhibit an unexpected esthetical and mathematical richness. There is some hope that they form a link to realistic, both invertible and continuous, systems; for example, in physics, chemistry, and technology.

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